Chapter 6

Why should ISRs be as brief as possible?

can mess with timing of other interrupts or itself.
could run so long where the ISR doesn't finish before
the next interrupt happens.

What is a safe way to send information from an ISR to a task in FreeRTOS?

using ISR variants of API's like:
xQueueSendFromISR()
xSemaphoreGiveFromISR()
vTaskNotifyGiveFromISR()

What is the purpose of a semaphore?

coordinate use of hardware between tasks. (traffic light).

Chapter 7

Describe the purpose of a mutex

protects shared resources by allowing ownership of the shared resource

What is priority inversion? How does priority inheritance solve this problem?

A high priority task can be blocked by a lower priority task because of the
mutex. allowing a different task to be ran than those two.

inheritance fixes this by letting the low priority have the high priority, which
then it can finish the task and release the mutex.

Chapter 8

What is the purpose of a FreeRTOS event group?

multiple events can be tracked in a single object. helps with synchronization

Chapter 9

Why are task notifications better than queues, semaphores, or event groups?

faster, low memory usage, and lightweight compared to the others.

What specific benefits are produced by the task notification strategy employed in the
UART example (compare Example 16 with Example 24)?

Lower Latency, Less overhead, more efficient, simpler.